

Network goodness calculus proposition

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Seminar

Part one

Abstract

- Environment - area with demand
- Network - satisfying demand
 - accessibility - only to use - Inertia - lack
 - gravitating - availability - environment zones
 - costs - length
- Network goodness - a number - the measure of
 - how the network is good to service the area that is embedded in.
- Network goodness - function
 - increasing - gravitating
 - decreasing - Inertia, length

Environment

Definition 1 (Network environment).

$\mathcal{U}$

- an administrative area - needs a service.
- $\mu(\mathcal{U})$ - the environment area size (km^2).

Definition 2 (Demand zone).

$\mathcal{T}_i, i = 1, \dots, n$ - a part of the network environment:

- $\mathcal{U} = \bigcup_{i=1}^n \mathcal{T}_i$
- $\mathcal{T}_i \cap \mathcal{T}_j = \emptyset$ for $i \neq j$.

A demand point

Definition 3.

$P_{ij}, j = 1, \dots, m(i)$

- $P_{ij} \in \mathcal{T}_i \subset \mathcal{U}$ - an environment point
- accomplished with a positive value $Q(P_{ij}) > 0$
- $Q(P_{ij})$ - demand concentrated at P_{ij} .
- $m(i)$ - number of demand points within zone $\mathcal{T}_i$.
- maths: for any $\mathcal{T}_i$ there exists $m(i) \in \mathbb{N}$
 - $\bar{\mathbb{N}} = \{1, 2, \dots, m(i)\}$
 - $p_i : \bar{\mathbb{N}} \rightarrow \mathcal{T}_i, p_i(j) = P_{ij}, p_i(j) \neq p_i(k)$ when $j \neq k$
 - $Q_i : p_i(\bar{\mathbb{N}}) \rightarrow \mathcal{T}_i, Q_i(P_{ij}) = Q_{ij}$.

A segment in coordinated network environment

Definition 4.

The environment $\mathcal{U}$ is coordinated if for any point $P \in \mathcal{U}$ there exists a uniquely defined pair of real numbers (x, y) . Denote $P = (x, y)$.

Theorem 5.

Any network environment can be coordinated.

Definition 6 (A segment).

A segment between the points $A, B \in \mathcal{U}$, with coordinates $A = (x_A, y_A)$ and $B = (x_B, y_B)$ is a set consisting of all the points on the straight line between the points:

$$\overline{AB} = \{P : P = (1 - t)A + tB, t \in [0, 1]\}. \quad (0.1)$$

A segment graph

Definition 7 (Well-known).

A graph $G = (N, E)$ is a pair consisting of a set N and $E \subset N \times N$.

Theorem 8.

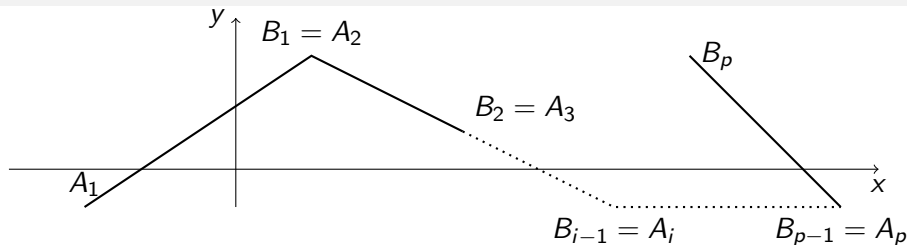
Any set of segments defines a graph.

Proof of Theorem 8.

- Given a set of segments $\{\overline{A_i B_i}, i = 1, 2, \dots, s\}$.
- Denote $\overline{A_i B_i} = (A_i, B_i)$ as element in $E = \{(A_i, B_i) \mid i = 1, 2, \dots, s\}$
- $|E| = s$.
- Any $\overline{A_i B_i}$ is giving two points into $N = \{A_i, B_i \mid i = 1, 2, \dots, s\}$
- $|N| \leq 2s$ elements.



A network is a connected graph



A path is $\{\overline{A_i B_i}, i = 1, \dots, p\}$ with $B_i = A_{i+1}$, $i = 1, \dots, p - 1$.

Definition 9 (Connected graph).

A graph $G = (N, E)$ is connected if any pair of nodes has a path that begins at one and ends at another node.

Definition 10 (Network).

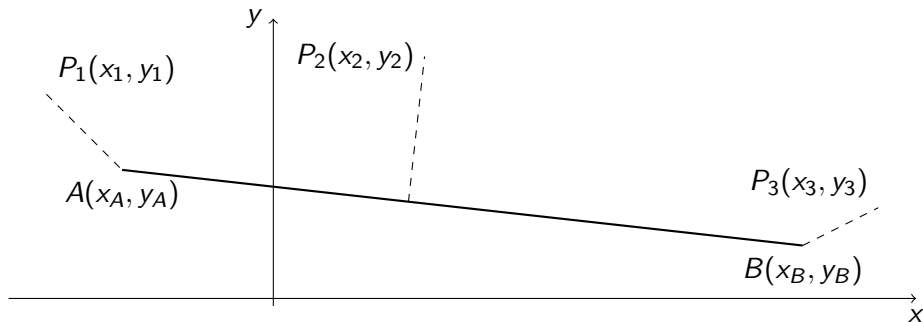
A network is a connected graph generated with a finite set of segments.

The segment equation

Theorem 11.

Let $A = (x_A, y_A)$ and $B = (x_B, y_B)$ be given points. The segment $\overline{AB}$ is defined as a set:

$$\overline{AB} = \{(x, y) \in [x_A, x_B] \times [y_A, y_B] : (y_B - y_A)x + (x_A - x_B)y + x_By_A - x_Ay_B = 0\}. \quad (0.2)$$



Square distance

Theorem 12.

Square distance between a point $P = (x, y)$ and a segment $\overline{AB}$ is given with

$$d^2(P, \overline{AB}) = (x - x_A)^2 + (y - y_A)^2, \text{ if} \\ (x - x_A)(x_B - x_A) + (y - y_A)(y_B - y_A) < 0 \quad (0.3)$$

$$(x - x_B)(x_A - x_B) + (y - y_B)(y_A - y_B) < 0, \quad (0.4)$$
$$d^2(P, \overline{AB}) = (x - x_B)^2 + (y - y_B)^2, \text{ if} \\ (x - x_B)(x_A - x_B) + (y - y_B)(y_A - y_B) < 0, \quad (0.5)$$

or

$$d^2(P, \overline{AB}) = \frac{[(y_B - y_A)x + (x_A - x_B)y + x_By_A - x_Ay_B]^2}{(x_A - x_B)^2 + (y_B - y_A)^2}, \quad (0.6)$$

in any other case.

A zone demand center

Theorem 13.

For each $\mathcal{T}_i$, there exists only one center point $P_i = (x_{\mathcal{T}_i}, y_{\mathcal{T}_i})$ with:

$$\begin{aligned} x_{\mathcal{T}_i} &= \frac{Q(P_{i1}) \cdot x_{i1} + Q(P_{i2}) \cdot x_{i2} + \cdots + Q(P_{im(i)}) \cdot x_{im(i)}}{Q(\mathcal{T}_i)} \\ y_{\mathcal{T}_i} &= \frac{Q(P_{i1}) \cdot y_{i1} + Q(P_{i2}) \cdot y_{i2} + \cdots + Q(P_{im(i)}) \cdot y_{im(i)}}{Q(\mathcal{T}_i)}, \quad (0.7) \end{aligned}$$

$Q(\mathcal{T}_i) = Q(P_{i1}) + Q(P_{i2}) + \cdots + Q(P_{im(i)})$ is a total zone's demand concentrated in P_i .

Dokaz.

A consequence from Theorem 5. □